Rattler Rec — Test Case Documentation

COP 3330 · Integration Project 1 · Florida A&M; University

*Complete the **Actual Output** and **P/F** columns by running each test before submitting. All 20 tests must be marked **P** (Pass). Submit this document as **test_cases.pdf** in your GitHub repository.*

TC # Test case ID	**Feature / Option** Which menu option or behavior is being tested	**Input / Steps** Exact inputs to enter or actions to take	**Expected Output** Key lines your terminal should show	**Actual Output** *Your output — fill in before submitting*	**P/F** Pass or Fail

Section A — Functional Test Cases (TC-01 through TC-13)

TC #	Feature / Option	Input / Steps	Expected Output	Actual Output (fill in)	P/F
TC-01	**Add Group Fitness** (Option 1)	ID: RR-GFC-001 Day: Monday, 10:00–11:30 Class: HBCU Zumba Instructor: INS-001 Dr. Maya Rivers Cert: Group, 8 yrs Difficulty: 3 Capacity: 20	"Session added successfully!" Option 4 shows: [GROUP FITNESS] RR-GFC-001 with all correct fields	Session added successfully! Select option: 4 [GROUP FITNESS] RR-GFC-001 Time : Monday 10:00-11:30 (90 min) Class : HBCU Zumba Difficulty : 3 / 5 Instructor : Dr. Maya Rivers Capacity : 0 / 20 enrolled	P
TC-02	**Add Personal Training** (Option 2)	ID: RR-PT-002 Day: Wednesday, 14:00–15:00 Trainer: INS-002 Coach Darius Webb Cert: Personal, 6 yrs Client: N01234567 Tiara Simmons, Student	"Session added successfully!" [PERSONAL TRAINING] RR-PT-002 Fee: $45.00 (= $15 + 6 × $5)	Drop-in Fee: 9.00 Session added successfully! [PERSONAL TRAINING] RR-PT-002 Time : Wednesday 14:00-15:00 (60 min) Trainer : Coach Darius Webb Client : Tiara Simmons [Student] Fee : $45.00	P
TC-03	**Add Open Gym** (Option 3)	ID: RR-OG-003 Day: Friday, 06:00–08:00 Area: Weights Monitored: No Capacity: 30	"Session added successfully!" [OPEN GYM] RR-OG-003 Drop-in Fee: $2.00	[OPEN GYM] RR-OG-003 Time : Friday 06:00-08:00 (120 min) Area : Weights Monitored : No Capacity : 0 / 30 enrolled Drop-in Fee: $2.00	P
TC-04	**View Full Schedule** (Option 4)	Prereq: TC-01, TC-02, TC-03 Select option 4	All 3 sessions display in order Type tags correct: [GROUP FITNESS] [PERSONAL TRAINING] [OPEN GYM]		P
TC-05	**Total Revenue** (Option 5)	Schedule: TC-01 + TC-02 + TC-03 Select option 5	"Total potential revenue: $56.00" ($9.00 + $45.00 + $2.00)	Total potential revenue: $56.00	P
TC-06	**Filter — Group Fitness** (Option 6)	Select option 6 Enter type: Group Fitness	Only TC-01 displayed No other sessions shown		P
TC-07	**Filter — Open Gym** (Option 6)	Select option 6 Enter type: Open Gym	Only TC-03 displayed No other sessions shown		P
TC-08	**Enroll in Session** (Option 7)	Select option 7 Session ID: RR-GFC-001	"Enrolled successfully. (1/20)" Option 4 confirms: Capacity: 1 / 20 enrolled	Enrolled successfully. (1/20)	P

TC #	Feature / Option	Input / Steps	Expected Output	Actual Output (fill in)	P/F
TC-09	**Drop from Session** (Option 8)	Prereq: TC-08 (1 enrolled) Select option 8 Session ID: RR-GFC-001	"Dropped successfully. (0/20)" Capacity returns to 0 / 20	Dropped successfully. (0/20)	P
TC-10	**Most Popular** (Option 9)	Enroll 3 times in RR-GFC-001 Enroll 1 time in RR-OG-003 Select option 9	Displays RR-GFC-001 (enrollment = 3, highest) All session fields shown correctly		P
TC-11	**Instructor Count** (Option 10)	After adding TC-01 and TC-02 (2 Instructor objects created) Select option 10	"Active instructors on record: 2" Static s_instructorCount = 2		P
TC-12	**Member Count** (Option 11)	After adding TC-02 (1 Member object created) Select option 11	"Active members on record: 1" Static s_memberCount = 1		P
TC-13	**TimeSlot format** in display output	Select option 4 after TC-01 Observe the Time line	Time line reads: Monday 10:00-11:30 (90 min) Correct day, HH:MM-HH:MM, duration		P

Section B — Edge Case & Error Handling Tests (TC-14 through TC-20)

TC #	Feature / Option	Input / Steps	Expected Output	Actual Output (fill in)	P/F
TC-14	**Invalid day name** (Option 1 or 3)	Enter day: Funday Then re-enter: Monday	Error rejecting "Funday" Re-prompt shown Session added on valid input		P
TC-15	**End time before start** (Option 1 or 3)	Start: 14:00 End: 13:00 Then re-enter valid end time	Error: end must be after start Re-prompt shown Session added after correction		P
TC-16	**Difficulty out of range** (Option 1)	Try: 0 Then: -1 Then: 6 Then valid: 4	Error for each invalid value Session added only on valid input		P
TC-17	**Enroll — unknown ID** (Option 7)	Select option 7 Session ID: RR-FAKE-999	"Session not found." Program does not crash Returns to main menu		P
TC-18	**Enroll past capacity** (Option 7)	Add Open Gym with capacity: 2 Enroll twice (both succeed) Attempt a third enrollment	Enrollments 1 and 2: success Enrollment 3: "Session is full." No crash		P
TC-19	**Drop with zero** enrollment (Option 8)	Session with 0 enrolled Select option 8 on that session	"No enrollments to drop." Enrollment stays at 0 No crash		P
TC-20	**Most popular —** empty schedule (Option 9)	No sessions added Select option 9	std::runtime_error caught Descriptive message printed Program continues running		P

Name: Kennedy Lewis Date: 4/12/26 GitHub Repo: git@github.com:Kennedy-Lewis/Rattler_Rec_Project1.git All tests passed: ■ Yes ■ No